III. Climbers (Vine)

An elongated weak stem generally supported by means of climbing devices are called **Climbers** (vines) which may be annual or perennial, herbaceous or woody.

Liana is a vine that is perennial and woody. Liana's are major components in the tree canopy layer of some tropical forests. Example: *Ventilago, Entada*, *Bougainvillea.*

IV. Trees

A tree is a stout, tall, perennial, woody plant having one main stem called **trunk** with many lateral branches. Example: Mango, Sapota, Jack, Fig, Teak. If the trunk remains unbranched it is said to be **caudex.** Example: Palmyra, Coconut.

3.2 Plant habitat

Depending upon where plants grow habitats may be classified into two major categories: I. Terrestrial and II.Aquatic.

I. Terrestrial

Plants growing on land are called **terrestrial plants.**

II. Aquatic

Plants that are living in water are called **aquatic plants** or hydrophytes.

3.3 Life Span

Based on life span plants are classified into 3 types. They are annuals, biennials and perennials

I. Annuals

A plant that completes its life cycle in one growing season. Example: Maize, Water melon, Groundnut, Rice.

II. Biennials

A plant that lives for two seasons, growing vegetatively during the first season and flowering and fruiting during the second season. Example: Carrot, Radish, Cabbage.

III. Perennials

A plant that grows for many years that flowers and set fruits for several seasons during the life span. When they bear fruits every year, they are called **polycarpic perennials.** Example: Mango, Sapota. Some plants produce flowers and fruits only once and die after a vegetative growth of several years. These plants are called **monocarpic perennials**. Example: *Bambusa*, *Agave, Musa.*

3.4 Parts of a flowering plant

Flowering plants are called **"Angiosperms"** or **Magnoliophytes**. They are sporophytes consisting of an axis with an underground **"Root system"** and an aerial **"Shoot System".** The shoot system has a stem, branches and leaves. The root system consists of root and its lateral branches.

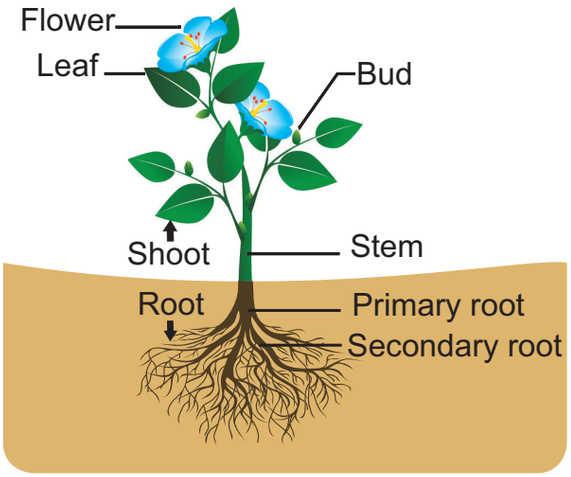


Figure: 3.1: Parts of a flowering plant

3.5 Root System

The root is non-green, cylindrical descending axis of the plant that usually grows into the soil (positively geotropic). It develops from the radicle which is the first structure that comes out when a seed is placed in the soil. Root is responsible for absorption of water and nutrients and anchoring the plant.

I. Characteristic features

- Root is the descending portion of the plant axis.
- Generally non-green in colour as it lacks chlorophyll.
- Does not possess nodes, internodes and buds (Exception in sweet potato and members of Rutaceae, roots bear buds which help in vegetative propagation)

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